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Stability of Peridynamic Correspondence Material Models and Their Particle Discretizations

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Abstract

Peridynamic correspondence material models provide a way to combine a material model from the local theory with the inherent capabilities of peridynamics to model long-range forces and fracture. However, correspondence models in a typical particle discretization suffer from zero-energy mode instability. These instabilities are shown here to be an aspect of material stability. A stability condition is derived for state-based materials starting from the requirement of potential energy minimization. It is shown that all correspondence materials fail this stability condition due to zero-energy deformation modes of the family. To eliminate these modes, a term is added to the correspondence strain energy density that resists deviations from a uniform deformation. The resulting material model satisfies the stability condition while effectively leaving the stress tensor unchanged. Computational examples demonstrate the effectiveness of the modified material model in avoiding zero-energy mode instability in a peridynamic particle code.

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1 Introduction

Peridynamics is a nonlocal theory of mechanics in which material points in a continuum or in a group of discrete particles interact with each other through forces. The exchange of forces between any pair of material points is called a *bond interaction*. These forces are determined by the material model according to the current deformation.

Material models in peridynamics explicitly determine the force in each bond. Many material models have been investigated as the peridynamic theory has been developed. A particularly useful class of material models is called the *correspondence* models. These models use a stress-strain relation from the local theory as a means to determine the bond forces. In principle, any material model from a finite element code can be applied directly in a peridynamic computational simulation when formulated as a correspondence model.

Correspondence material models have been used successfully in the modeling of a wide variety of material response, including elastic-plastic and viscoplastic materials with large strains and fracture, for example [6, 5, 21]. However, the usefulness of peridynamic correspondence models has been impeded by some practical issues in their implementation in meshless peridynamic codes. One such issue is the difficulty that is sometimes encountered when evaluating the *shape tensor* defined in equation (15) below, particularly when very large deformations or extensive damage are present. A more troublesome issue with correspondence models is their tendency to exhibit zero-energy mode oscillations in meshless simulations. These oscillations typically build up gradually during a long simulation, but they can eventually ruin the results. They can occur even when the appropriate version [16] of the Courant-Friedrichs-Levy time step restriction is applied. Zero-energy mode instabilities can occur in a variety of particle methods as well as finite elements, motivating the development of corrective techniques [4, 20, 9]. The purpose of the present paper is to investigate the root cause of this type of instability as a manifestation of a *material* instability, rather than merely an artifact of the meshless discretization. A stabilized correspondence material model is proposed that is stable in both the continuum and discretized equations.

2 Peridynamics background

Peridynamics is a generalization of the standard theory of solid mechanics. It treats the mechanics of continuous bodies and discrete particles, including long-range forces and fracture, within the same basic field equations. A typical material point $\mathbf{x}$ in a body $\mathcal{B}$ interacts with other material points $\mathbf{q}$ within a neighborhood called the *family* of $\mathbf{x}$, denoted $\mathcal{H}$. The radius of the neighborhood is called the *horizon*, δ. The vector in the undeformed configuration from $\mathbf{x}$ to any of its neighbors $\mathbf{q} \in \mathcal{H}$ is called a *bond*, generically given the symbol $\boldsymbol{\xi} = \mathbf{q} - \mathbf{x}$. The vector $\boldsymbol{\xi} = \mathbf{0}$ is excluded from the family.

The equation of motion in peridynamics is given by

$$\rho(\mathbf{x})\ddot{\mathbf{y}}(\mathbf{x},t) = \int_{\mathcal{H}} \mathbf{f}(\mathbf{q},\mathbf{x},t)\, \mathrm{d}V_{\mathbf{q}} + \mathbf{b}(\mathbf{x},t) \quad \forall \mathbf{x} \in \mathcal{B},\ t \geq 0, \tag{1}$$

where ρ is the density, $\mathbf{y}$ is the deformation, and $\mathbf{b}$ is the prescribed body force density field. The equation of equilibrium (see equation (24) below for the derivation) is written

$$\int_{\mathcal{H}} \mathbf{f}(\mathbf{q},\mathbf{x})\, \mathrm{d}V_{\mathbf{q}} + \mathbf{b}(\mathbf{x}) = \mathbf{0} \quad \forall \mathbf{x} \in \mathcal{B}. \tag{2}$$

The vector-valued function $\mathbf{f}(\mathbf{q},\mathbf{x},t)$ is the *pairwise bond force density* (with units of force per volume squared) that $\mathbf{q}$ exerts on $\mathbf{x}$. The material model determines the values of $\mathbf{f}$ according to the deformations of the families of $\mathbf{x}$ and of $\mathbf{q}$.

The simplest class of material model [14] is called *bond-based*, in which $\mathbf{f}(\mathbf{q},\mathbf{x},t)$ depends *only* on the deformation of the bond $\mathbf{q} - \mathbf{x}$. More general material response can be incorporated using the *state-based* class of models [17], in which $\mathbf{f}(\mathbf{q},\mathbf{x},t)$ depends not only on $\mathbf{q} - \mathbf{x}$, but also on the deformation of all the other bonds in $\mathcal{H}$. State-based material models are expressed using mathematical objects called *states*, which are mappings from $\mathcal{H}$ to some other quantity, either vector- or scalar-valued. By convention, the bond that a state $\underline{\mathbf{A}}$ operates on is written in angle brackets, $\underline{\mathbf{A}}\langle\boldsymbol{\xi}\rangle$. Dependence of the state on position and time is written in square brackets, $\underline{\mathbf{A}}[\mathbf{x},t]$.

The *deformation state* $\underline{\mathbf{Y}}$ maps bonds onto their deformed images:

$$\underline{\mathbf{Y}}[\mathbf{x},t]\langle\boldsymbol{\xi}\rangle = \mathbf{y}(\mathbf{x}+\boldsymbol{\xi},t) - \mathbf{y}(\mathbf{x},t). \tag{3}$$

The bond force density $\mathbf{t}(\mathbf{q},\mathbf{x},t)$ that $\mathbf{q}$ exerts on $\mathbf{x}$ due to the material model at $\mathbf{x}$ is expressed in the *force state* $\underline{\mathbf{T}}$,

$$\underline{\mathbf{T}}[\mathbf{x},t]\langle\mathbf{q}-\mathbf{x}\rangle = \mathbf{t}(\mathbf{q},\mathbf{x},t). \tag{4}$$

Due to the requirement for balance of linear momentum, the pairwise bond force density contains contributions from the material models at both $\mathbf{x}$ and $\mathbf{q}$:

$$\begin{aligned} \mathbf{f}(\mathbf{q},\mathbf{x},t) &= \mathbf{t}(\mathbf{q},\mathbf{x},t) - \mathbf{t}(\mathbf{x},\mathbf{q},t) \\ &= \underline{\mathbf{T}}[\mathbf{x},t]\langle\mathbf{q}-\mathbf{x}\rangle - \underline{\mathbf{T}}[\mathbf{q},t]\langle\mathbf{x}-\mathbf{q}\rangle. \end{aligned} \tag{5}$$

A *material model* $\hat{\underline{\mathbf{T}}}$ is a function that maps a deformation state onto a force state:

$$\underline{\mathbf{T}}[\mathbf{x},t] = \hat{\underline{\mathbf{T}}}(\underline{\mathbf{Y}}[\mathbf{x},t]).$$

For example, a bond-based material model might have the form

$$\underline{\hat{\mathbf{T}}}\langle\boldsymbol{\xi}\rangle = \frac{1}{2}C(\boldsymbol{\xi})\underline{\mathbf{M}}\langle\boldsymbol{\xi}\rangle\frac{|\underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle| - |\boldsymbol{\xi}|}{|\boldsymbol{\xi}|}, \qquad \underline{\mathbf{M}}\langle\boldsymbol{\xi}\rangle = \frac{\underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle}{|\underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle|}, \tag{6}$$

where C is the symmetric *micromodulus* function, which is calibrated from the measured mechanical properties of the material, and $\underline{\mathbf{M}}$ is the *deformed bond direction state*.

The *dot product* of two vector states $\underline{\mathbf{A}}$ and $\underline{\mathbf{B}}$ is defined by

$$\underline{\mathbf{A}} \bullet \underline{\mathbf{B}} = \int_{\mathcal{H}} \underline{\mathbf{A}}\langle\boldsymbol{\xi}\rangle \cdot \underline{\mathbf{B}}\langle\boldsymbol{\xi}\rangle \, \mathrm{d}V_{\boldsymbol{\xi}},$$

where the small dot $(\cdot)$ signifies the usual inner product of two vectors in $\mathbb{R}^3$. Also define the norm of a vector state by

$$\|\underline{\mathbf{A}}\| = \sqrt{\underline{\mathbf{A}} \bullet \underline{\mathbf{A}}} \, .$$

If $\underline{\mathbf{A}}$ is a vector state and $\underline{v}$ is a scalar state, the product $\underline{v}\underline{\mathbf{A}}$ is a vector state defined by

$$(\underline{v}\underline{\mathbf{A}})\langle\boldsymbol{\xi}\rangle = \underline{v}\langle\boldsymbol{\xi}\rangle\underline{\mathbf{A}}\langle\boldsymbol{\xi}\rangle.$$

A notion of differentiation with respect to states will now be defined. Let Ψ be a scalar-valued function of a vector state $\underline{\mathbf{A}}$. Let $d\underline{\mathbf{A}}$ be a small increment in $\underline{\mathbf{A}}$. Suppose there is a vector state denoted $\Psi_{\underline{\mathbf{A}}}$ such that

$$\Psi(\underline{\mathbf{A}} + d\underline{\mathbf{A}}) = \Psi(\underline{\mathbf{A}}) + \Psi_{\underline{\mathbf{A}}} \bullet d\underline{\mathbf{A}} + o(\|d\underline{\mathbf{A}}\|) \qquad \forall d\underline{\mathbf{A}}. \tag{7}$$

Then $\Psi_{\underline{\mathbf{A}}}$ is called the *Fréchet derivative* of Ψ with respect to the state $\underline{\mathbf{A}}$.

Fréchet derivatives are an essential tool in peridynamic material modeling because of their role in elastic materials. In an elastic material, there is a *strain energy density function* $\hat{W}(\underline{\mathbf{Y}})$ that has the same mechanical meaning (reversibly stored energy per unit volume) as in the local theory. The force state in a peridynamic elastic material is found from

$$\underline{\hat{\mathbf{T}}}(\underline{\mathbf{Y}}) = \hat{W}_{\underline{\mathbf{Y}}}(\underline{\mathbf{Y}}). \tag{8}$$

From (7), this means that the increment in strain energy density due to an incremental change in the deformation state $d\underline{\mathbf{Y}}$ is given by

$$dW = \underline{\mathbf{T}} \bullet d\underline{\mathbf{Y}} = \int_{\mathcal{H}} \underline{\mathbf{T}}\langle\boldsymbol{\xi}\rangle \cdot d\underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle \, \mathrm{d}V_{\boldsymbol{\xi}}. \tag{9}$$

Mechanically, this integral sums up the incremental work done on all the bonds in the family as they deform, acting against the individual bond forces.

Given a force state, it is often useful to define a stress tensor that represents an approximate way of characterizing the internal forces in the local theory. One way to define this approximate stress tensor is called the *partial stress tensor*, denoted $\boldsymbol{\nu}$:

$$\boldsymbol{\nu} = \int_{\mathcal{H}} \underline{\mathbf{T}}\langle\boldsymbol{\xi}\rangle \otimes \boldsymbol{\xi} \, \mathrm{d}V_{\boldsymbol{\xi}}. \tag{10}$$

The partial stress tensor has the same properties, in terms of symmetries and invariances, as the first Piola stress. The properties of the partial stress tensor are described in [13].

A *double state* $\underline{\underline{\mathbb{C}}}$ is a tensor-valued function that operates on two bonds $\boldsymbol{\xi}, \boldsymbol{\zeta} \in \mathcal{H}$:

$$\underline{\mathbf{A}} \bullet \underline{\underline{\mathbb{C}}} \bullet \underline{\mathbf{B}} = \int_{\mathcal{H}} \int_{\mathcal{H}} \underline{\mathbf{A}}\langle \boldsymbol{\xi} \rangle \cdot \left(\underline{\underline{\mathbb{C}}}\langle \boldsymbol{\xi}, \boldsymbol{\zeta} \rangle \underline{\mathbf{B}}\langle \boldsymbol{\zeta} \rangle \right) \, \mathrm{d}V_{\boldsymbol{\zeta}} \, \mathrm{d}V_{\boldsymbol{\xi}}.$$

The second Fréchet derivative of the function Ψ, denoted $\Psi_{\underline{\mathbf{A}}\underline{\mathbf{A}}}$, is an example of a double state. Further information on the properties of states and operations on them can be found in [17]. Double states are discussed further in [15]. Extensive accounts of peridynamics are available in the review article [18] and the book by Madenci and Oterkus [12].

Because equations (1) and (2) do not involve partial derivatives of the deformation with respect to the spatial coordinates, they apply to continuous media, continuous media with evolving discontinuities (cracks), and discrete particles. Damage is usually modeled by allowing bonds to "break" irreversibly according to some failure criterion, after which the bond no longer sustains force. Discrete particles are usually treated using generalized functions in the material model, as indicated in equation (12) below.

3 Particle discretizations as peridynamic bodies

A particle discretization of the peridynamic equations was proposed in [16]. A body is discretized into nodes with undeformed positions $\{\mathbf{x}_i\}$ and undeformed volumes $\{V_i\}$. The equilibrium equation (2) is approximated by

$$\sum_{j\in\mathcal{H}_i} \mathbf{f}(\mathbf{x}_j, \mathbf{x}_i)V_j + \mathbf{b}(\mathbf{x}_i) = \mathbf{0} \tag{11}$$

for each node i. The $\mathbf{f}$ in (11) is identical to the one in the original continuum equation (2). This discretization has been the workhorse in many applications of peridynamic mechanics, although others are possible and have certain advantages.

A unique property of the discretized peridynamic equation (11), unlike other discretizations of peridynamics and all discretizations of the local equations, is that it is, in a certain sense, exact. It is not an exact approximation to the continuum equations. However, it is an exact model of a different peridynamic body consisting of a set of discrete points. To see this, define the following body and material model:

$$\begin{aligned} \tilde{\mathcal{B}} &= \{\mathbf{x}_i\}, \\ \tilde{\mathbf{f}}(\mathbf{q}, \mathbf{x}) &= \sum_{i\in\tilde{\mathcal{B}}}\sum_{j\in\tilde{\mathcal{B}}} \mathbf{f}(\mathbf{x}_j, \mathbf{x}_i)\Delta(\mathbf{q}-\mathbf{x}_j)\Delta(\mathbf{x}-\mathbf{x}_i)V_iV_j, \\ \tilde{\mathbf{b}}(\mathbf{x}) &= \sum_{i\in\tilde{\mathcal{B}}} \mathbf{b}(\mathbf{x}_i)\Delta(\mathbf{x}-\mathbf{x}_i)V_i, \end{aligned} \tag{12}$$

where Δ is the Dirac delta function. (Note that the dimensions of Δ are 1/volume.) It may be confirmed by substituting (12) into the original equilibrium equation (2) that it is equivalent to the discretized continuum model in (11).

The significance of this observation is that a particle discretization is *in itself* a legitimate peridynamic body, namely $\tilde{\mathcal{B}}$, with pairwise force density given by $\tilde{\mathbf{f}}$. This means that any results about peridynamic bodies immediately apply to the numerical model without further analysis or adaptation. This fact will be used later in this paper to show that zero-energy modes in a discretized model can be understood as an aspect of material stability. In particular, if a material model is shown to be stable, this guarantees that the discretized model will also be stable. (This applies only to the spatial discretization. Time discretization of the dynamic equations still involves restrictions on the time step size if a conditionally stable time integration method is used. It is also assumed that the discretized model has a sufficient number of points and that they are arranged such that the shape tensor $\mathbf{K}$ is nonsingular.)

4 Correspondence material models

Correspondence materials [17] use a stress tensor as an intermediate step in determining the bond forces. Suppose a local material model is given of the form

$$\boldsymbol{\sigma}(\mathbf{x}) = \hat{\boldsymbol{\sigma}}(\mathbf{F}(\mathbf{x})), \qquad \mathbf{F}(\mathbf{x}) = \frac{\partial \mathbf{y}}{\partial \mathbf{x}}(\mathbf{x}), \tag{13}$$

where $\boldsymbol{\sigma}$ is the first Piola stress and $\mathbf{F}$ is the deformation gradient tensor. In peridynamic correspondence material models, $\mathbf{F}$ is found from an approximate expression:

$$\bar{\mathbf{F}} = \left(\int_{\mathcal{H}} \underline{\omega}\langle \boldsymbol{\xi} \rangle \underline{\mathbf{Y}}\langle \boldsymbol{\xi} \rangle \otimes \boldsymbol{\xi} \, \mathrm{d}V_{\boldsymbol{\xi}} \right) \mathbf{K}^{-1}, \tag{14}$$

where $\underline{\omega}$ is a positive-valued weighting function called the *influence state*, and $\mathbf{K}$ is the symmetric positive definite *shape tensor* defined by

$$\mathbf{K} = \int_{\mathcal{H}} \underline{\omega}\langle \boldsymbol{\xi} \rangle \boldsymbol{\xi} \otimes \boldsymbol{\xi} \, \mathrm{d}V_{\boldsymbol{\xi}}. \tag{15}$$

The approximate deformation gradient is exact if the deformation is uniform (linear): if there is a tensor $\mathbf{F}_0$ such that $\underline{\mathbf{Y}}\langle \boldsymbol{\xi} \rangle = \mathbf{F}_0 \boldsymbol{\xi}$ for all $\boldsymbol{\xi} \in \mathcal{H}$, then

$$\begin{aligned} \bar{\mathbf{F}} &= \left(\int_{\mathcal{H}} \underline{\omega}\langle \boldsymbol{\xi} \rangle \mathbf{F}_0 \boldsymbol{\xi} \otimes \boldsymbol{\xi} \, \mathrm{d}V_{\boldsymbol{\xi}} \right) \mathbf{K}^{-1} \\ &= \mathbf{F}_0 \left(\int_{\mathcal{H}} \underline{\omega}\langle \boldsymbol{\xi} \rangle \boldsymbol{\xi} \otimes \boldsymbol{\xi} \, \mathrm{d}V_{\boldsymbol{\xi}} \right) \mathbf{K}^{-1} \\ &= \mathbf{F}_0. \end{aligned} \tag{16}$$

The local material model $\hat{\boldsymbol{\sigma}}$ can include other dependences such as time, rate, and history.

The force state in a correspondence material [17] is given by

$$\underline{\mathbf{T}}^c\langle \boldsymbol{\xi} \rangle = \underline{\omega}\langle \boldsymbol{\xi} \rangle \hat{\boldsymbol{\sigma}}(\bar{\mathbf{F}}) \mathbf{K}^{-1} \boldsymbol{\xi}. \tag{17}$$

It is easily confirmed from (10) and (17) that the partial stress tensor for a peridynamic correspondence material reproduces the underlying local material model:

$$\boldsymbol{\nu}^c = \hat{\boldsymbol{\sigma}}(\bar{\mathbf{F}}) \mathbf{K}^{-1} \int_{\mathcal{H}} \underline{\omega}\langle \boldsymbol{\xi} \rangle \boldsymbol{\xi} \otimes \boldsymbol{\xi} \, \mathrm{d}V_{\boldsymbol{\xi}} = \hat{\boldsymbol{\sigma}}(\bar{\mathbf{F}}), \tag{18}$$

where (15) has been used. (Unlike (16), the identity (18) holds regardless of whether the deformation is uniform.)

Suppose the underlying local material model is elastic, and let the local strain energy density function be $\hat{W}$:

$$\hat{\boldsymbol{\sigma}} = \frac{\partial \hat{W}}{\partial \mathbf{F}}.$$

Then the peridynamic correspondence material derived from this local material model is also elastic [17]:

$$\underline{\hat{\mathbf{T}}}^c = \hat{W}^c_{\underline{\mathbf{Y}}}, \qquad \hat{W}^c(\underline{\mathbf{Y}}) = \hat{W}(\bar{\mathbf{F}}(\underline{\mathbf{Y}})), \tag{19}$$

where now the Fréchet derivative is used rather than the tensor gradient.

As mentioned in the Introduction, computational simulations using correspondence material models are susceptible to weak instabilities due to zero-energy ocsillatory modes. These have been reported in several publications and are commonly seen in practice. Ganzenmüller, Hiermaier, and May [7] interpret the zero-energy modes in meshless simulations as an aspect of the numerical discretization, drawing on similarities with rank deficiency effects in Smoothed-Particle Hydrodynamics (SPH). Closely related results are discussed by Bessa, Foster, Belytschko, and Liu [1].

Numerical stabilization techniques are commonly used with correspondence material models to suppress the zero-energy modes. Littlewood [10, 11] introduced corrective forces on the bonds that tend to drive nodes in a meshless model toward smooth deformations. Breitenfeld, Geubelle, Weckner, and Silling [2] proposed stabilizing forces that push a material particle closer to the geometrical center of its deformed family. Damping forces and artificial viscosity are also somewhat effective in reducing zero-energy modes [13]. Tupek and Radovitzky [19] investigated the use of alternative strain measures to help avoid the closely related phenomenon of interpenetration in meshless peridynamic grids.

Up to now, the phenomenon of zero-energy modes in correspondence materials has generally been viewed as a consequence of meshless discretization. Instead, the purpose of the present work is to study the root cause of this phenomenon as a *material instability* rather than a *numerical instability* [8]. To pursue this, a general criterion for material stability in peridynamics will first be discussed.

5 Material stability and energy minimization

One concept of material stability is to require that in equilibrium, total potential energy of a bounded elastic body must, in addition to being stationary with respect to the first variation of the deformation, also be a local minimum. To investigate the implications of this requirement for material models, define the total potential energy of a bounded elastic peridynamic body by

$$\Pi(\mathbf{y}) = \int_{\mathcal{B}} (\hat{W}(\underline{\mathbf{Y}}[\mathbf{x}]) - \mathbf{b}\cdot\mathbf{y})\, \mathrm{d}V_{\mathbf{x}}. \tag{20}$$

Here, $\mathbf{y} \in \mathcal{E}(\mathcal{B})$, where $\mathcal{E}(\mathcal{B})$ is the set of deformations of $\mathcal{B}$ satisfying any essential (prescribed displacement) boundary conditions, and excluding rigid deformations. Define the Gâteaux derivative of $\Pi(\mathbf{y})$ in the direction of a given displacement field $\mathbf{u}$ by

$$D_{\mathbf{u}}\Pi(\mathbf{y}) = \frac{d\Pi(\mathbf{y}+\epsilon\mathbf{u})}{d\epsilon}, \qquad \mathbf{y}+\epsilon\mathbf{u} \in \mathcal{E}(\mathcal{B}). \tag{21}$$

Let $\underline{\mathbf{U}}$ denote the displacement state corresponding to $\mathbf{u}$,

$$\underline{\mathbf{U}}[\mathbf{x}]\langle \mathbf{q}-\mathbf{x}\rangle = \mathbf{u}(\mathbf{q}) - \mathbf{u}(\mathbf{x}). \tag{22}$$

If Π is stationary at $\mathbf{y}$, then

$$D_{\mathbf{u}}\Pi(\mathbf{y}) = 0 \tag{23}$$

for all admissible displacement fields $\mathbf{u}$. Then from (8), (20), (22), and (23), for any $\mathbf{u}$,

$$\begin{aligned}
D_{\mathbf{u}}\Pi(\mathbf{y}) &= \int_{\mathcal{B}} (\hat{W}_{\underline{\mathbf{Y}}}(\underline{\mathbf{Y}}) \bullet \underline{\mathbf{U}} - \mathbf{b}\cdot\mathbf{u})\, \mathrm{d}V_{\mathbf{x}} \\
&= \int_{\mathcal{B}} (\underline{\hat{\mathbf{T}}}(\underline{\mathbf{Y}}) \bullet \underline{\mathbf{U}} - \mathbf{b}\cdot\mathbf{u})\, \mathrm{d}V_{\mathbf{x}} \\
&= \int_{\mathcal{B}} \left(\int_{\mathcal{B}} \underline{\mathbf{T}}[\mathbf{x}]\langle \mathbf{q}-\mathbf{x}\rangle \cdot (\mathbf{u}(\mathbf{q}) - \mathbf{u}(\mathbf{x}))\, \mathrm{d}V_{\mathbf{q}} - \mathbf{b}(\mathbf{x})\cdot\mathbf{u}(\mathbf{x}) \right) \mathrm{d}V_{\mathbf{x}} \\
&= \int_{\mathcal{B}} \left(\int_{\mathcal{B}} (\underline{\mathbf{T}}[\mathbf{q}]\langle \mathbf{x}-\mathbf{q}\rangle - \underline{\mathbf{T}}[\mathbf{x}]\langle \mathbf{q}-\mathbf{x}\rangle)\, \mathrm{d}V_{\mathbf{q}} - \mathbf{b}(\mathbf{x}) \right) \cdot \mathbf{u}(\mathbf{x})\, \mathrm{d}V_{\mathbf{x}},
\end{aligned} \tag{24}$$

where the change of variables $\mathbf{x} \leftrightarrow \mathbf{q}$ is used in the last step. Localization leads to the conclusion that $D_{\mathbf{u}}\Pi(\mathbf{y}) = 0$ implies

$$\int_{\mathcal{H}} (\underline{\mathbf{T}}[\mathbf{x}]\langle \mathbf{q}-\mathbf{x}\rangle - \underline{\mathbf{T}}[\mathbf{q}]\langle \mathbf{x}-\mathbf{q}\rangle)\, \mathrm{d}V_{\mathbf{q}} + \mathbf{b}(\mathbf{x}) = \mathbf{0} \quad \forall \mathbf{x} \in \mathcal{B}, \tag{25}$$

which is the peridynamic equation of equilibrium (2). (25) is a condition on the deformation $\mathbf{y}$, since $\underline{\mathbf{T}} = \underline{\hat{\mathbf{T}}}(\underline{\mathbf{Y}})$. Working backwards shows that (25) is sufficient as well as necessary for stationary Π.

Define the second Gâteaux derivative by

$$D^2_{\mathbf{u}}\Pi(\mathbf{y}) = \frac{d^2\Pi(\mathbf{y}+\epsilon\mathbf{u})}{d\epsilon^2} \tag{26}$$

for admissible displacement fields $\mathbf{u}$. Returning to (24), one finds that

$$D^2_{\mathbf{u}}\Pi(\mathbf{y}) = \int_{\mathcal{B}} \underline{\mathbf{U}} \bullet \underline{\underline{\mathbb{K}}}(\underline{\mathbf{Y}}) \bullet \underline{\mathbf{U}} \, \mathrm{d}V_{\mathbf{x}}, \tag{27}$$

where $\underline{\underline{\mathbb{K}}}$ is the *micromodulus double state* [15] defined by

$$\underline{\underline{\mathbb{K}}} = \underline{\underline{\hat{\mathbf{T}}}}_{\underline{\mathbf{Y}}} = \hat{W}_{\underline{\mathbf{Y}}\underline{\mathbf{Y}}}.$$

Suppose Π is stationary at a deformation $\mathbf{y}$; therefore (23) and (25) hold. If, additionally, $\mathbf{y}$ is a local minimizer of Π, then for any admissible $\mathbf{u}$,

$$D^2_{\mathbf{u}}\Pi(\mathbf{y}) > 0. \tag{28}$$

Then from (27) and (28),

$$\int_{\mathcal{B}} \underline{\mathbf{U}} \bullet \underline{\underline{\mathbb{K}}} \bullet \underline{\mathbf{U}} \, \mathrm{d}V_{\mathbf{x}} > 0. \tag{29}$$

To avoid the need to evaluate $\underline{\underline{\mathbb{K}}}$ explicitly, it is convenient to rewrite this expression in the more suggestive notation

$$\int_{\mathcal{B}} d\underline{\mathbf{T}} \bullet d\underline{\mathbf{Y}} \, \mathrm{d}V_{\mathbf{x}} > 0, \tag{30}$$

for any admissible $d\mathbf{y}$, where $d\underline{\mathbf{T}}$ is the change in force state resulting from an incremental change in the deformation state $d\underline{\mathbf{Y}}$:

$$d\underline{\mathbf{T}} = \underline{\underline{\mathbb{K}}}(\underline{\mathbf{Y}}) \bullet d\underline{\mathbf{Y}}.$$

A sufficient condition for (30) to hold is clearly that

$$d\underline{\mathbf{T}} \bullet d\underline{\mathbf{Y}} > 0 \qquad \text{for all admissible } d\underline{\mathbf{Y}}. \tag{31}$$

To investigate the status of correspondence materials with respect to the stability condition (31), it is helpful to establish the following connection between the stability of the peridynamic and underlying local material models.

Proposition 1. Let a correspondence material model $\underline{\hat{\mathbf{T}}}^c(\underline{\mathbf{Y}})$ be given, with underlying local material model $\hat{\boldsymbol{\sigma}}(\mathbf{F})$. Let $d\underline{\mathbf{Y}}$ be any admissible increment in $\underline{\mathbf{Y}}$. Then

$$d\underline{\mathbf{T}}^c \bullet d\underline{\mathbf{Y}} = d\boldsymbol{\sigma} \cdot d\bar{\mathbf{F}}. \tag{32}$$

Proof. Write (17) in component form (the underscores are omitted from symbols with indices to improve readability):

$$T^c_i\langle\boldsymbol{\xi}\rangle = \underline{\omega}\langle\boldsymbol{\xi}\rangle\hat{\sigma}_{ij}(\bar{\mathbf{F}})K^{-1}_{jk}\xi_k, \tag{33}$$

thus

$$dT^c_i\langle\boldsymbol{\xi}\rangle = \underline{\omega}\langle\boldsymbol{\xi}\rangle(c_{ijmn}d\bar{F}_{mn})K^{-1}_{jk}\xi_k, \tag{34}$$

where the c_{ijmn} are the components of the fourth order elasticity tensor. From (14), for a given incremental deformation state $d\underline{\mathbf{Y}}$,

$$d\bar{F}_{mn} = \int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\zeta}\rangle dY_m\langle\boldsymbol{\zeta}\rangle\zeta_\ell K^{-1}_{\ell n} \, \mathrm{d}V_{\boldsymbol{\zeta}}. \tag{35}$$

Combining (34) and (35),

$$\begin{aligned} d\underline{\mathbf{T}}^c \bullet d\underline{\mathbf{Y}} &= \int_{\mathcal{H}} dT_i\langle\boldsymbol{\xi}\rangle dY_i\langle\boldsymbol{\xi}\rangle \, \mathrm{d}V_{\boldsymbol{\xi}} \\ &= \int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle c_{ijmn} \left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\zeta}\rangle dY_m\langle\boldsymbol{\zeta}\rangle \zeta_\ell K^{-1}_{\ell n} \, \mathrm{d}V_{\boldsymbol{\zeta}} \right) K^{-1}_{jk} \xi_k dY_i\langle\boldsymbol{\xi}\rangle \, \mathrm{d}V_{\boldsymbol{\xi}} \\ &= c_{ijmn} \left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\zeta}\rangle dY_m\langle\boldsymbol{\zeta}\rangle \zeta_\ell K^{-1}_{\ell n} \, \mathrm{d}V_{\boldsymbol{\zeta}} \right) \left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle dY_i\langle\boldsymbol{\xi}\rangle \xi_k K^{-1}_{jk} \, \mathrm{d}V_{\boldsymbol{\xi}} \right) \\ &= c_{ijmn} d\bar{F}_{mn} d\bar{F}_{ij} \\ &= d\sigma_{ij} d\bar{F}_{ij}, \end{aligned}$$

which proves (32). □

For the remainder of this discussion, it will be assumed that the underlying local material model is stable:

$$d\boldsymbol{\sigma} \cdot d\mathbf{F} > 0 \tag{36}$$

for all $d\mathbf{F}$ other than rigid rotations. This assumption excludes softening types of physical material response. The following result shows that all correspondence materials fail the stability condition derived above.

Proposition 2. Let a correspondence material model $\underline{\mathbf{T}} = \hat{\underline{\mathbf{T}}}^c(\underline{\mathbf{Y}})$ be given, with underlying local material model $\hat{\boldsymbol{\sigma}}(\mathbf{F})$. Then the stability condition (31) fails to hold.

Proof. Consider an incremental deformation state $d\underline{\mathbf{Y}}$ such that

$$\underline{\omega}\langle -\boldsymbol{\xi}\rangle d\underline{\mathbf{Y}}\langle -\boldsymbol{\xi}\rangle = \underline{\omega}\langle\boldsymbol{\xi}\rangle d\underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle \qquad \forall \boldsymbol{\xi} \in \mathcal{H}.$$

Then from (14), $d\bar{\mathbf{F}} = \mathbf{0}$. Therefore, by Proposition 1, $d\underline{\mathbf{T}} \bullet d\underline{\mathbf{Y}} = 0$, proving the result. □.

Proposition 2 shows that $\bar{\mathbf{F}}$ is invariant under a large class of deformations of $\mathcal{H}$. Similar conclusions have been obtained by Lehoucq [8] by investigating the null space of the operator in (14), using the formalism of nonlocal vector calculus [3]. Such cases of admissible $d\underline{\mathbf{Y}}$ such that $d\bar{\mathbf{F}} = \mathbf{0}$ can reasonably called *zero-energy modes* of deformation of the family, because they fail to change the potential energy. In discretized models, zero-energy modes of deformation are bad news because they manifest themselves as numerical instabilities. This type of instability is a source of practical difficulty in finite elements and particle discretizations of both the local PDEs and the peridynamic equations when using correspondence material models.

6 Stabilized correspondence material models

In the previous section, it was shown that there are increments $d\underline{\mathbf{Y}}$ that have no effect on the approximate deformation gradient $\bar{\mathbf{F}}$, leading to zero-energy models of deformation. Here, we propose to add a term to the strain energy density that resists these modes of deformation.

Define the *nonuniform part* of the deformation state $\underline{\mathbf{z}}$ by

$$\underline{\mathbf{z}}\langle\boldsymbol{\xi}\rangle = \underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle - \bar{\mathbf{F}}\boldsymbol{\xi}, \tag{37}$$

where $\bar{\mathbf{F}}$ is given by (14). Observe from (14) and (37) that the approximate deformation gradient operator applied to the nonuniform part of the deformation state vanishes:

$$\begin{aligned}
&\left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle\underline{\mathbf{z}}\langle\boldsymbol{\xi}\rangle \otimes \boldsymbol{\xi}\, \mathrm{d}V_{\boldsymbol{\xi}}\right)\mathbf{K}^{-1} \\
&\quad= \left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle(\underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle - \bar{\mathbf{F}}\boldsymbol{\xi}) \otimes \boldsymbol{\xi}\, \mathrm{d}V_{\boldsymbol{\xi}}\right)\mathbf{K}^{-1} \\
&\quad= \left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle\underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle \otimes \boldsymbol{\xi}\, \mathrm{d}V_{\boldsymbol{\xi}}\right)\mathbf{K}^{-1} - \bar{\mathbf{F}}\left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle\boldsymbol{\xi} \otimes \boldsymbol{\xi}\, \mathrm{d}V_{\boldsymbol{\xi}}\right)\mathbf{K}^{-1} \\
&\quad= \bar{\mathbf{F}} - \bar{\mathbf{F}}\mathbf{K}\mathbf{K}^{-1} \\
&\quad= \mathbf{0}.
\end{aligned} \tag{38}$$

The nonuniform part of the deformation state contains that portion of the deformation state that deviates from a uniform deformation of the family.

Consider an elastic material with strain energy density given by

$$\hat{W} = \hat{W}^c + \hat{W}^s, \tag{39}$$

where $\hat{W}^c$ is the energy from the correspondence model (19) and

$$\hat{W}^s(\underline{\mathbf{Y}}) = \frac{1}{2}(\underline{\beta}\underline{\mathbf{z}}) \bullet \underline{\mathbf{z}}, \tag{40}$$

where $\underline{\beta}$ is a prescribed positive-valued scalar state.

Now we evaluate the force state corresponding to $\hat{W}^s$. For any small $d\underline{\mathbf{Y}}$, it follows from (14), (37), and (40) that

$$\begin{aligned}
dW^c &= (\underline{\beta}\underline{\mathbf{z}}) \bullet d\underline{\mathbf{z}} \\
&= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle(dY_i\langle\boldsymbol{\xi}\rangle - d\bar{F}_{ij}\xi_j)\, \mathrm{d}V_{\boldsymbol{\xi}} \\
&= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle dY_i\langle\boldsymbol{\xi}\rangle\, \mathrm{d}V_{\boldsymbol{\xi}} - \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle z_i\langle\boldsymbol{\eta}\rangle d\bar{F}_{ij}\eta_j\, \mathrm{d}V_{\boldsymbol{\eta}} \\
&= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle dY_i\langle\xi\rangle\, \mathrm{d}V_{\boldsymbol{\xi}} \\
&\qquad - \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle z_i\langle\boldsymbol{\eta}\rangle\eta_j\, \mathrm{d}V_{\boldsymbol{\eta}}\right)\left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle dY_i\langle\boldsymbol{\xi}\rangle\xi_p\, \mathrm{d}V_{\boldsymbol{\xi}}\right)K^{-1}_{pj} \\
&= \int_{\mathcal{H}} \left[\underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle - \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle z_i\langle\boldsymbol{\eta}\rangle\eta_j\, \mathrm{d}V_{\boldsymbol{\eta}}\right)\underline{\omega}\langle\boldsymbol{\xi}\rangle\xi_p K^{-1}_{pj}\right] dY_i\langle\boldsymbol{\xi}\rangle\, \mathrm{d}V_{\boldsymbol{\xi}}.
\end{aligned}$$

Comparing the last expression with (7) and (8), evidently

$$T_i^s\langle\boldsymbol{\xi}\rangle = \underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle - \underline{\omega}\langle\boldsymbol{\xi}\rangle \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle z_i\langle\boldsymbol{\eta}\rangle\eta_j \, \mathrm{d}V_{\boldsymbol{\eta}}\right) K_{pj}^{-1}\xi_p \tag{41}$$

or

$$\underline{\mathbf{T}}^s\langle\boldsymbol{\xi}\rangle = \underline{\beta}\langle\boldsymbol{\xi}\rangle\underline{\mathbf{z}}\langle\boldsymbol{\xi}\rangle - \underline{\omega}\langle\boldsymbol{\xi}\rangle \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle\underline{\mathbf{z}}\langle\boldsymbol{\eta}\rangle \otimes \boldsymbol{\eta} \, \mathrm{d}V_{\boldsymbol{\eta}}\right) \mathbf{K}^{-1}\boldsymbol{\xi}. \tag{42}$$

Proposition 3. Let $\underline{\mathbf{T}}^s$ be given by (41) or equivalently by (42). Let $\boldsymbol{\nu}^s$ be the partial stress for $\underline{\mathbf{T}}^s$. Then

$$\boldsymbol{\nu}^s = \mathbf{0}. \tag{43}$$

Proof. By (10) and (41),

$$\begin{aligned}
\nu_{ij}^s &= \int_{\mathcal{H}} T_i^s\langle\boldsymbol{\xi}\rangle\xi_j \, \mathrm{d}V_{\boldsymbol{\xi}} \\
&= \int_{\mathcal{H}} \left[\underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle - \underline{\omega}\langle\boldsymbol{\xi}\rangle \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle z_i\langle\boldsymbol{\eta}\rangle\eta_k \, \mathrm{d}V_{\boldsymbol{\eta}}\right) K_{pk}^{-1}\xi_p\right] \xi_j \, \mathrm{d}V_{\boldsymbol{\xi}} \\
&= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle\xi_j \, \mathrm{d}V_{\boldsymbol{\xi}} - \left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle\xi_p\xi_j \, \mathrm{d}V_{\boldsymbol{\xi}}\right) K_{pk}^{-1} \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle z_i\langle\boldsymbol{\eta}\rangle\eta_k \, \mathrm{d}V_{\boldsymbol{\eta}}\right) \\
&= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle\xi_j \, \mathrm{d}V_{\boldsymbol{\xi}} - K_{jp}K_{pk}^{-1} \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle z_i\langle\boldsymbol{\eta}\rangle\eta_k \, \mathrm{d}V_{\boldsymbol{\eta}}\right) \\
&= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle z_i\langle\boldsymbol{\xi}\rangle\xi_j \, \mathrm{d}V_{\boldsymbol{\xi}} - \delta_{jk} \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle z_i\langle\boldsymbol{\eta}\rangle\eta_k \, \mathrm{d}V_{\boldsymbol{\eta}}\right) \\
&= 0.
\end{aligned}$$

□

Proposition 4. Let $\underline{\mathbf{T}}^s$ be given by (41) or equivalently by (42). Let $d\underline{\mathbf{Y}}$ be an admissible nonzero increment in the deformation state. Then

$$d\underline{\mathbf{T}}^s \bullet d\underline{\mathbf{Y}} \geq 0. \tag{44}$$

Moreover, strict inequality holds if the nonuniform part of the deformation state is non-zero:

$$d\underline{\mathbf{z}} \neq \underline{\mathbf{0}} \quad \Longrightarrow \quad d\underline{\mathbf{T}}^s \bullet d\underline{\mathbf{Y}} > 0. \tag{45}$$

Proof. By (41),

$$\begin{aligned} d\underline{\mathbf{T}}^s \bullet d\underline{\mathbf{Y}} &= \int_{\mathcal{H}} dY_i\langle\boldsymbol{\xi}\rangle \left[\underline{\beta}\langle\boldsymbol{\xi}\rangle dz_i\langle\boldsymbol{\xi}\rangle - \underline{\omega}\langle\boldsymbol{\xi}\rangle \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle dz_i\langle\boldsymbol{\eta}\rangle\eta_j \, \mathrm{d}V_{\boldsymbol{\eta}}\right) K^{-1}_{pj}\xi_p\right] \mathrm{d}V_{\boldsymbol{\xi}} \\ &= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle dY_i\langle\boldsymbol{\xi}\rangle dz_i\langle\boldsymbol{\xi}\rangle \, \mathrm{d}V_{\boldsymbol{\xi}} \\ &\quad - \left(\int_{\mathcal{H}} \underline{\omega}\langle\boldsymbol{\xi}\rangle dY_i\langle\boldsymbol{\xi}\rangle\xi_p \, \mathrm{d}V_{\boldsymbol{\xi}}\right) K^{-1}_{pj} \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle dz_i\langle\boldsymbol{\eta}\rangle\eta_j \, \mathrm{d}V_{\boldsymbol{\eta}}\right) \\ &= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle dY_i\langle\boldsymbol{\xi}\rangle dz_i\langle\boldsymbol{\xi}\rangle \, \mathrm{d}V_{\boldsymbol{\xi}} - d\bar{F}_{ij} \left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle dz_i\langle\boldsymbol{\eta}\rangle\eta_j \, \mathrm{d}V_{\boldsymbol{\eta}}\right) \\ &= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle dz_i\langle\boldsymbol{\xi}\rangle \Big(dY_i\langle\boldsymbol{\xi}\rangle - d\bar{F}_{ij}\xi_j\Big) \, \mathrm{d}V_{\boldsymbol{\xi}} \\ &= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle dz_i\langle\boldsymbol{\xi}\rangle dz_i\langle\boldsymbol{\xi}\rangle \, \mathrm{d}V_{\boldsymbol{\xi}} \\ &= \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\xi}\rangle |d\mathbf{z}\langle\boldsymbol{\xi}\rangle|^2, \end{aligned} \tag{46}$$

proving (44) and, since $\underline{\beta} > 0$, also (45). □

So far we have shown that the new strain energy density term defined by (40) is does not alter the stress tensor, is stable in the sense of (45), and generates a force state through its Fréchet derivative given by (41) or (42). The non-polarity of $\hat{\underline{\mathbf{T}}}^s$ follows from the fact that is elastic and objective (because it has no preferred directions) [15].

Now we return to the full material model whose strain energy density is given by (39). Since

$$\hat{\underline{\mathbf{T}}} = \hat{\underline{\mathbf{T}}}^c + \hat{\underline{\mathbf{T}}}^s, \tag{47}$$

it follows from Proposition 3 that

$$\boldsymbol{\nu} = \boldsymbol{\nu}^c + \boldsymbol{\nu}^s = \hat{\boldsymbol{\sigma}}, \tag{48}$$

in other words the partial stress is left unchanged by the new term $\hat{W}^s$. The only remaining question is whether the combined material model (40) is stable.

Proposition 5. Let $\hat{\mathbf{T}}$ be given by (47), and let $d\underline{\mathbf{Y}}$ be any admissible nonzero increment in the deformation state. Then the stability condition (31) holds.

Proof. By (47) and Proposition 1,

$$\begin{aligned} d\underline{\mathbf{T}} \bullet d\underline{\mathbf{Y}} &= d\underline{\mathbf{T}}^c \bullet d\underline{\mathbf{Y}} + d\underline{\mathbf{T}}^s \bullet d\underline{\mathbf{Y}} \\ &= d\boldsymbol{\sigma} \bullet d\bar{\mathbf{F}} + d\underline{\mathbf{T}}^s \bullet d\underline{\mathbf{Y}}. \end{aligned} \tag{49}$$

There are only two possibilities:

- $d\bar{\mathbf{F}} \neq \mathbf{0}$. In this case, (36) and (44) imply that

$$d\boldsymbol{\sigma} \bullet d\bar{\mathbf{F}} > 0 \quad \text{and} \quad d\underline{\mathbf{T}}^s \bullet d\underline{\mathbf{Y}} \geq 0.$$

- $d\bar{\mathbf{F}} = \mathbf{0}$. In this case, it follows from (37) that $\underline{\mathbf{z}} \neq \underline{\mathbf{0}}$. Therefore, from (45),

$$d\boldsymbol{\sigma} \bullet d\bar{\mathbf{F}} = 0 \quad \text{and} \quad d\underline{\mathbf{T}}^s \bullet d\underline{\mathbf{Y}} > 0.$$

In either case, (49) proves that

$$d\underline{\mathbf{T}} \bullet d\underline{\mathbf{Y}} > 0. \tag{50}$$

□

From (17), (42) and (47), the *stabilized* material model (39) has a force state given by

$$\begin{aligned} \hat{\underline{\mathbf{T}}}\langle\boldsymbol{\xi}\rangle &= \hat{\underline{\mathbf{T}}}^c + \hat{\underline{\mathbf{T}}}^s \\ &= \underline{\omega}\langle\boldsymbol{\xi}\rangle\hat{\boldsymbol{\sigma}}(\bar{\mathbf{F}})\mathbf{K}^{-1}\boldsymbol{\xi} \\ &\quad + \underline{\beta}\langle\boldsymbol{\xi}\rangle\underline{\mathbf{z}}\langle\boldsymbol{\xi}\rangle - \underline{\omega}\langle\boldsymbol{\xi}\rangle\left(\int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle\underline{\mathbf{z}}\langle\boldsymbol{\eta}\rangle \otimes \boldsymbol{\eta}\, \mathrm{d}V_{\boldsymbol{\eta}}\right)\mathbf{K}^{-1}\boldsymbol{\xi}, \end{aligned} \tag{51}$$

where $\underline{\mathbf{z}}$ is given by (37). A slightly more suggestive form of (51) is given by

$$\hat{\underline{\mathbf{T}}}\langle\boldsymbol{\xi}\rangle = \underline{\omega}\langle\boldsymbol{\xi}\rangle(\hat{\boldsymbol{\sigma}}(\bar{\mathbf{F}}) - \boldsymbol{\sigma}^s)\mathbf{K}^{-1}\boldsymbol{\xi} + \underline{\beta}\langle\boldsymbol{\xi}\rangle\underline{\mathbf{z}}\langle\boldsymbol{\xi}\rangle, \tag{52}$$

where $\boldsymbol{\sigma}^s$ is defined by

$$\boldsymbol{\sigma}^s = \int_{\mathcal{H}} \underline{\beta}\langle\boldsymbol{\eta}\rangle\underline{\mathbf{z}}\langle\boldsymbol{\eta}\rangle \otimes \boldsymbol{\eta}\, \mathrm{d}V_{\boldsymbol{\eta}}. \tag{53}$$

In view of (38), the particular choice of $\underline{\beta}$ given by

$$\underline{\beta} = \gamma\underline{\omega}, \tag{54}$$

where γ is a positive constant, results in a further simplification of (52):

$$\hat{\underline{\mathbf{T}}}\langle\boldsymbol{\xi}\rangle = \underline{\omega}\langle\boldsymbol{\xi}\rangle\big(\hat{\boldsymbol{\sigma}}(\bar{\mathbf{F}})\mathbf{K}^{-1}\boldsymbol{\xi} + \gamma\underline{\mathbf{z}}\langle\boldsymbol{\xi}\rangle\big). \tag{55}$$

The assumption (54) may or may not be the best choice for an application, particularly when extensive damage in a material is anticipated. Typically, in practice, damage in a bond $\boldsymbol{\xi}$ is modeled by setting $\underline{\omega}\langle\boldsymbol{\xi}\rangle = 0$ ("bond breakage"). Under the assumption (54), such bond breakage would also be applied to the stabilization term $\underline{\mathbf{T}}^s\langle\boldsymbol{\xi}\rangle$. This would remove any interaction between the material points $\mathbf{x}$ and $\mathbf{x} + \boldsymbol{\xi}$ (except possibly from the material model at $\mathbf{x} + \boldsymbol{\xi}$) and therefore tend to promote the "dust" effect that is often seen in computational simulations.

The appropriate value of γ can be estimated from the elastic material properties. Observing that the stabilization term in (55) resembles a bond-based material response, γ can be chosen so that this term contains some fraction α of the strain energy density in a simple deformation such as isotropic expansion with linear strain ϵ. The energy contribution from the stabilization term is estimated from

$$W^s \approx \frac{1}{4}\int_{\mathcal{H}} \gamma\underline{\omega}\langle\boldsymbol{\xi}\rangle(\epsilon|\boldsymbol{\xi}|)^2(4\pi|\boldsymbol{\xi}|^2)\, \mathrm{d}V_{\boldsymbol{\xi}} = \alpha k(3\epsilon)^2,$$

where k is the bulk modulus. In three dimensions, assuming that $\underline{\omega}$ depends only on $|\boldsymbol{\xi}|$, the resulting estimate for γ is found to be

$$\gamma = \frac{9k\alpha}{\pi\int_0^\delta \omega(r)r^4\, dr}. \tag{56}$$

The stabilization method proposed above is conceptually similar to the method proposed by Littlewood [10], which also applies forces that tend to reduce deviations of $\underline{\mathbf{Y}}$ from $\bar{\mathbf{F}}$. Littlewood's method, in the present notation, introduces statilization forces of the form

$$\underline{\mathbf{T}}^s\langle\boldsymbol{\xi}\rangle = A\underline{\mathbf{M}}\langle\boldsymbol{\xi}\rangle\frac{|\underline{\mathbf{Y}}\langle\boldsymbol{\xi}\rangle| - |\bar{\mathbf{F}}\boldsymbol{\xi}|}{|\boldsymbol{\xi}|},$$

where A is a constant and $\underline{\mathbf{M}}\langle\boldsymbol{\xi}\rangle$ is the deformed bond direction. Littlewood also introduced a modified but essentially similar form of the above [11]. Neither method has not been analyzed with respect to its effect on material stability or the partial stress tensor as described above.

7 Examples

Three examples are presented here. All three model an aluminum coupon with dimensions 150mm $\times$ 50mm $\times$ 5mm that contains a hole with diameter 20mm. It is modeled using a peridynamic correspondence material model. The grid spacing is 1mm. Other than the modification to the material model described in the previous sections, these calculations did not use damping or other stabilization forces. The particle discretization and explicit time integration method are described in [16].

In the first example, the plate is modeled as linear elastic. Figure 1 compares the velocity contours on the midplane of the coupon after being stretched at constant rate to a nominal tensile strain of 0.02%. The velocity fields with and without stabilization of the correspondence material model are shown. The zero-energy mode oscillations are evident in the left plot, without the stabilization. The stabilization term is obtained from (55) with $\alpha = 1$. The zero-energy mode oscillations are apparently eliminated when the stabilized material model is used, as shown in the right plot.

In the second example, the plate is modeled as a brittle linear elastic material with a bond failure strain of 0.2%. Bond breakage is implemented by setting the influence function $\underline{\omega}\langle\boldsymbol{\xi}\rangle$ to zero when the bond $\boldsymbol{\xi}$ breaks. Cracks nucleate at the stress concentrations near the hole and immediately grow toward the free edges of the specimen. The calculation uses the stabilized correspondence material model with $\alpha = 1$. Figure 2 shows the deformed grid and contours of damage after failure of the coupon. There is no evidence of instability in the result, and apparently the stabilization forces do not interfere with the progression of damage.

In the third example, the same coupon is modeled as elastic-plastic using the peridynamic implementation of plasticity described in [21]. The flow stress includes the effect of strain hardening using the Johnson-Cook plasticity model. Stabilization is used as described above. Necking occurs near the hole in the coupon. This simulation demonstrates that physically reasonable results without zero-energy mode instability are obtained using the stabilization method, even with fairly large plastic strain. The results are shown in Figure 3.

8 Conclusions

It was shown in this paper that any correspondence material model $\underline{\hat{\mathbf{T}}}^c$ is always unstable in the sense of (31). The instability manifests itself as zero-energy mode oscillations in a particle discretization.

However, the modified material model in $\underline{\hat{\mathbf{T}}}$ given by (47) is stable. From the discussion in Section 3, this result applies equally to continuum and discrete models. The stabilized correspondence material model satisfies the requirements of peridynamic material models (objectivity and non-polarity) and is elastic. The stabilization forces do not change the partial stress tensor. The new terms require negligible computer resources. The numerical examples suggest that this approach to stabilization is compatible with elastic and inelastic correspondence models and with bond breakage as a means to incorporate damage.

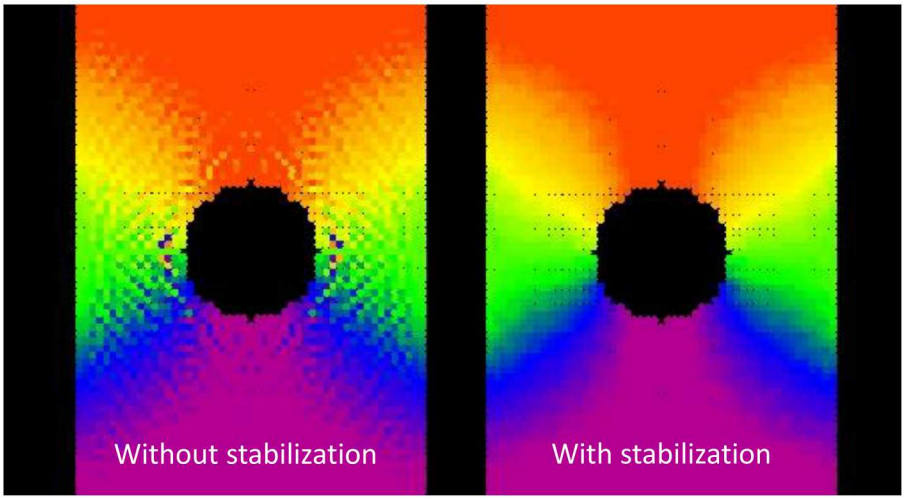


Figure 1. Contours of velocity in simulations of a linear elastic coupon under tension. Left: without stabilization, showing zero-energy mode instability. Right: with stabilization.

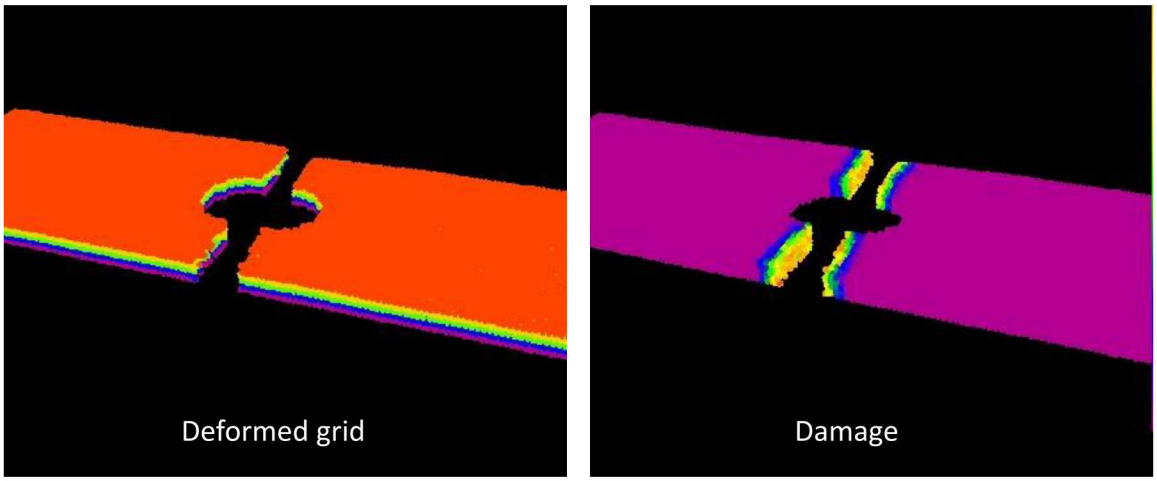


Figure 2. Simulation of a brittle linear elastic coupon under tension with stabilization forces. Left: grid configuration after failure. Right: contours of damage.

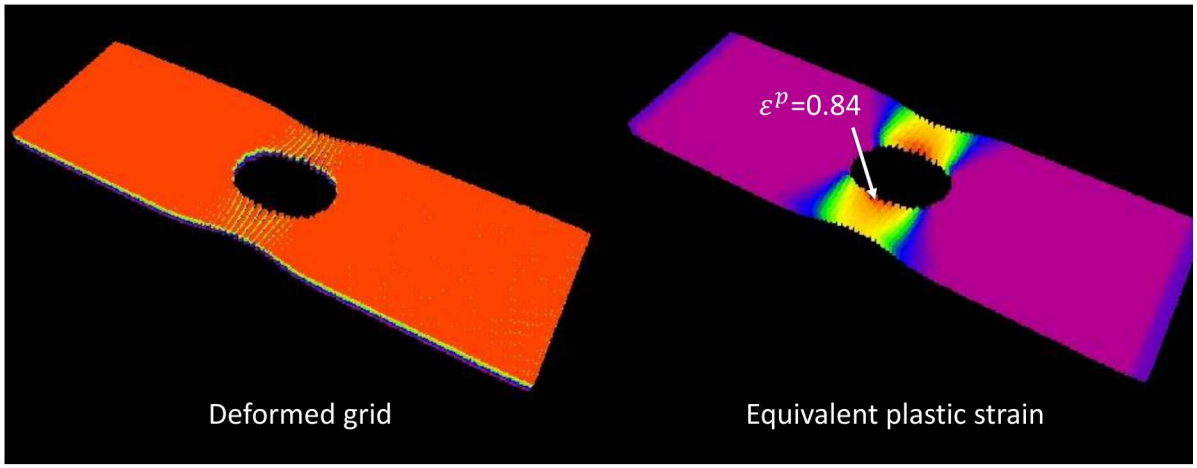


Figure 3. Simulation of a ductile coupon under tension with stabilization forces illustrating necking. Left: grid configuration at a global strain of 9%. Right: contours of equivalent plastic strain ε^p.

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